

installation-guide

BobaLink Browser Extension — Installation Guide

Document Version: 1.0.0

Last Updated: 2026-06-06

Audience: System Administrators, End Users

Companion Documents: [User Guide](#) | [Technical Specification](#) | [Developer Guide](#)

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1. Prerequisites

1.1 Infrastructure Requirements

Before installing BobaLink, ensure the following backend infrastructure is in place:

Option A: Boba Project (Recommended)

The Boba orchestration server provides the full feature set including search aggregation, SSE progress streaming, and auto-discovery.

Component	Requirement	Verification Command
Docker Engine	24.0+	<code>docker --version</code>
Docker Compose	2.20+	<code>docker compose version</code>
Available ports	8443 (Boba HTTPS)	<code>netstat -tlnp grep 8443</code>
SSL certificate	Valid or self-signed	<code>curl -k https://localhost:8443/health</code>

Quick verification that Boba is running:

```
# Using curl (replace with your Boba server IP)
curl -k https://192.168.1.100:8443/api/v1/health
```

```
# Expected response:
# {"status":"healthy","version":"1.4.2",...}
```

If Boba is not yet installed, refer to the Boba Project documentation for Docker Compose setup before proceeding.

Option B: qBitTorrent WebUI (Direct Mode)

For direct connection without the Boba middleware:

Component	Requirement	Verification
qBitTorrent	4.4.0 or later	Check in qBitTorrent → Help → About
WebUI enabled	Configured and running	Access <code>https://your-qbt:8080</code>
HTTPS access	TLS certificate (self-signed OK)	Browser can reach the URL
Authentication	Username/password configured	Login page accessible

Enabling qBitTorrent WebUI:

1. Open qBitTorrent on your server.

2. Navigate to **Tools → Options** (or **Edit → Preferences** on Linux).
3. Select **Web UI** from the left sidebar.
4. Check “**Web User Interface (Remote control)**”.
5. Configure the following:

Setting	Recommended Value	Description
IP Address	* (all interfaces) or specific IP	Which network interface to bind
Port	8080	WebUI port (change if needed)
Use HTTPS	Enabled	Required for secure communication
Username	admin (change this)	WebUI login username
Password	Strong unique password	WebUI login password
Bypass authentication for localhost	Disabled	Enforce auth on all connections

1. If using HTTPS (recommended), configure the certificate:
 - **Option 1:** Use qBitTorrent’s built-in self-signed certificate.
 - **Option 2:** Provide your own certificate and private key files.
2. Click **Apply** or **Save**.
3. Restart qBitTorrent to apply changes.

Verify WebUI accessibility:

```
# Test from the same machine
curl -k -X POST "https://localhost:8080/api/v2/auth/login" \
  -H "Referer: https://localhost:8080" \
  -d "username=admin" \
  -d "password=yourpassword"

# Expected: "Ok.SID" response with Set-Cookie header
```

1.2 Browser Requirements

Browser	Minimum Version	Download URL
Google Chrome	88	https://www.google.com/chrome/

Browser	Minimum Version	Download URL
Mozilla Firefox	109	https://www.mozilla.org/firefox/
Opera	74	https://www.opera.com/download
Yandex Browser	21	https://browser.yandex.com/
Microsoft Edge	88	https://www.microsoft.com/edge

Check your browser version:

Browser	Method
Chrome	Menu (:) → Help → About Google Chrome
Firefox	Menu (≡) → Help → About Firefox
Opera	Menu → Help → About Opera
Yandex	Menu → Advanced → About Yandex Browser
Edge	Menu (...) → Help and feedback → About Microsoft Edge

1.3 Network Requirements

Requirement	Description
Same network (recommended)	Browser and server on the same LAN for lowest latency
HTTPS accessibility	Browser must be able to reach the server over HTTPS
Certificate trust	Self-signed certificates may require browser exceptions
Firewall rules	Ports 8443 (Boba) and/or 8080 (qBitTorrent) must be open
No proxy interference	Corporate proxies may need exceptions for local addresses

Testing network connectivity:

```
# Replace with your server IP
SERVER_IP="192.168.1.100"
```

```
# Test basic connectivity
ping -c 3 $SERVER_IP
```

```
# Test port reachability (Boba)
nc -zv $SERVER_IP 8443

# Test port reachability (qBitTorrent)
nc -zv $SERVER_IP 8080

# Test HTTPS endpoint (accepts self-signed certs)
curl -k -I https://$SERVER_IP:8443/api/v1/health

# Test from the browser's context
# Open DevTools Console (F12) and run:
# fetch('https://192.168.1.100:8443/api/v1/health', {mode: 'no-
#   cors'})
# .then(r => console.log('Reachable'))
# .catch(e => console.error('Not reachable:', e));
```

2. Browser-Specific Installation

2.1 Chrome (Chrome Web Store)

The recommended installation method for Chrome users.

Step-by-step:

1. Open Google Chrome.
2. Navigate to the Chrome Web Store page for BobaLink:
 - Search “BobaLink” in the Chrome Web Store, OR
 - Direct link: <https://chrome.google.com/webstore/detail/bobalink/{extension-id}>
3. Click the **“Add to Chrome”** button.
4. Review the permission request:

Permission	Why It’s Needed
Read and change data on websites	Detect magnet links on web pages
Storage	Save configuration and queue locally
Context menus	Add “Send to Boba” right-click options
Notifications	Show download status toasts

1. Click **“Add extension”** to confirm.

2. The BobaLink icon (tea bubble) appears in the toolbar.

If the icon is hidden: 1. Click the extensions icon (puzzle piece) in the Chrome toolbar. 2. Find BobaLink in the list. 3. Click the pin icon to keep it visible in the toolbar.

2.2 Chrome (Manual Installation)

For environments where the Chrome Web Store is unavailable (enterprise, air-gapped networks).

Prerequisites: - BobaLink release ZIP file (boba-link-chrome-v1.0.0.zip) - Chrome Developer Mode enabled

Step-by-step:

1. Download the release ZIP from GitHub Releases:

```
wget https://github.com/bobaproject/boba-link/releases/download/v1.0.0/boba-link-chrome-v1.0.0.zip
```

2. Extract the ZIP to a permanent location:

```
unzip boba-link-chrome-v1.0.0.zip -d /opt/boba-link/chrome
```

Important: Do not delete this folder while the extension is installed. Chrome reads from this directory.

3. Open Chrome and navigate to `chrome://extensions`.

4. Enable “**Developer mode**” using the toggle in the top-right corner.

5. Click “**Load unpacked**”.

6. Select the extracted folder (`/opt/boba-link/chrome`).

7. BobaLink appears in the extensions list with its extension ID.

8. (Optional) Note the extension ID for debugging:

Extension ID: `abcdefghijklmnopqrstuvwxyzabcdef` (example)

2.3 Firefox (Add-ons Mozilla.org — AMO)

Step-by-step:

1. Open Mozilla Firefox.

2. Navigate to `https://addons.mozilla.org/en-US/firefox/addon/boba-link/`.

3. Click **“Add to Firefox”**.
4. Review the permission dialog:

Permission	Why It’s Needed
Access browser tabs	Enumerate tab groups for batch sending
Access browser storage	Save settings and queue
Access your data for all websites	Detect magnet links
Display notifications	Show send status

1. Click **“Add”** to confirm.
2. BobaLink is installed and the icon appears in the toolbar.

2.4 Firefox (Manual Installation)

For environments without AMO access.

Step-by-step:

1. Download the Firefox XPI or ZIP from GitHub Releases:

```
wget https://github.com/bobaproject/bobalink/releases/download/v1.0.0/bobalink-firefox-v1.0.0.zip
```

2. Extract to a permanent location:

```
unzip bobalink-firefox-v1.0.0.zip -d /opt/bobalink/firefox
```

3. Open Firefox and navigate to `about:debugging`.
4. Click **“This Firefox”** on the left sidebar.
5. Click **“Load Temporary Add-on...”**.
6. Navigate to the extracted folder and select `manifest.json`.
7. BobaLink loads as a temporary extension.

Note: Temporary extensions are removed when Firefox closes. For permanent manual installation on Firefox, the extension must be signed by Mozilla or installed via enterprise policies.

Firefox Enterprise Policy Installation (for managed environments):

Create or edit the policies file:

```
// Linux: /etc/firefox/policies/policies.json
// Windows: C:\Program Files\Mozilla
//           Firefox\distribution\policies.json
// macOS: /Applications/Firefox.app/Contents/Resources/
//         distribution/policies.json

{
  "policies": {
    "Extensions": {
      "Install": [
        "https://your-server/bobalink-firefox-v1.0.0-signed.xpi"
      ],
      "Locked": false
    }
  }
}
```

2.5 Opera (Opera Addons)

1. Open Opera Browser.
2. Navigate to <https://addons.opera.com/en/extensions/details/bobalink/>.
3. Click **“Add to Opera”**.
4. Confirm the installation when prompted.
5. The icon appears in the Opera toolbar.

2.6 Opera (Manual Installation)

Opera is Chromium-based and can install Chrome extensions.

1. Enable the “Install Chrome Extensions” feature:
 - Go to `opera://extensions`.
 - Enable **“Developer mode”**.
2. Download the Chrome release ZIP (same as Chrome manual installation).
3. Extract and load unpacked as described in the Chrome manual section.
4. Alternatively, install the Chrome Web Store version:
 - Install the “Install Chrome Extensions” addon from Opera Addons.
 - Visit the BobaLink Chrome Web Store page.

- Click “Add to Opera”.

2.7 Yandex Browser (Manual Installation)

Yandex Browser does not have a dedicated extension store for BobaLink. Use manual installation.

Step-by-step:

1. Download the Chrome release ZIP from GitHub.
2. Extract to a permanent location.
3. Open Yandex Browser and navigate to `browser://extensions`.
4. Enable “**Developer mode**”.
5. Click “**Load unpacked**”.
6. Select the extracted Chrome build folder.
7. BobaLink installs and the icon appears in the toolbar.

2.8 Chromium (Developer Mode)

For ungoogled-chromium and other Chromium derivatives:

1. Download the Chrome release ZIP.
2. Extract to a permanent location.
3. Navigate to `chrome://extensions`.
4. Enable “**Developer mode**”.
5. Click “**Load unpacked**” and select the extracted folder.

Command-line installation (advanced):

```
# Launch Chromium with the extension pre-loaded  
chromium --load-extension=/opt/bobalink/chrome \  
         --enable-logging=stderr \  
         --v=1
```

3. Manual Installation (All Browsers)

This section provides a unified manual installation procedure that works across all supported browsers.

3.1 Download Release Package

1. Visit the GitHub Releases page: <https://github.com/bobaproject/bobalink/releases>

2. Download the appropriate ZIP for your browser:

Browser	File
Chrome / Opera / Yandex / Edge	bobalink-chrome-v{version}.zip
Firefox	bobalink-firefox-v{version}.zip

1. Verify the download (optional but recommended):

```
# Download the checksum file
wget https://github.com/bobaproject/bobalink/releases/download/
v1.0.0/SHA256SUMS

# Verify
sha256sum -c SHA256SUMS
# Expected: bobalink-chrome-v1.0.0.zip: OK
```

3.2 Extract Archive

```
# Create installation directory
sudo mkdir -p /opt/bobalink

# Extract Chrome build
sudo unzip bobalink-chrome-v1.0.0.zip -d /opt/bobalink/chrome

# Or extract Firefox build
sudo unzip bobalink-firefox-v1.0.0.zip -d /opt/bobalink/firefox

# Set appropriate permissions
sudo chown -R $USER:$USER /opt/bobalink
sudo chmod -R 755 /opt/bobalink
```

Windows users: Use PowerShell Expand-Archive or 7-Zip to extract.

3.3 Load in Browser

Browser	Navigation	Action
Chrome	chrome://extensions	Enable Developer mode → Load unpacked
Firefox	about:debugging → This Firefox	Load Temporary Add-on → Select manifest.json
Opera	opera://extensions	

Browser	Navigation	Action
		Enable Developer mode → Load unpacked
Yandex	browser://extensions	Enable Developer mode → Load unpacked
Edge	edge://extensions	Enable Developer mode → Load unpacked

3.4 Post-Load Verification

After loading: 1. The extension should appear in the extensions list with version 1.0.0. 2. No error messages should appear. 3. The BobaLink icon should be visible in the browser toolbar.

4. Post-Installation Setup

4.1 Open Options Page

Immediately after installation, configure BobaLink:

1. Click the BobaLink icon in the toolbar.
2. Click the **gear icon** () in the popup.
3. The Options page opens in a new tab.

Alternatively: - Right-click the BobaLink icon → “**Options**”. - Or navigate to your browser’s extensions page and click “**Options**” under BobaLink.

4.2 Configure Boba Server URL

For Boba Server mode:

1. In Options → Server Settings, ensure “**Boba Server**” is selected as the connection mode.
2. Enter your Boba server URL:
 - Docker on same machine: `https://localhost:8443`
 - Docker on LAN server: `https://192.168.1.100:8443`
 - With custom port: `https://your-server:9000`

3. Select **“API Key”** as the authentication method.
4. Enter your API key.
5. Click **“Test Connection”**.
6. Wait for the green **“Connected”** indicator.

For qBitTorrent Direct mode:

1. In Options → Server Settings, select **“qBitTorrent Direct”**.
2. Enter your qBitTorrent WebUI URL:
 - Same machine: `https://localhost:8080`
 - LAN server: `https://192.168.1.100:8080`
3. Enter your WebUI username and password.
4. Click **“Test Connection”**.
5. Wait for the green **“Connected”** indicator.

4.3 Run Connection Test

The connection test verifies: - Network reachability (TCP connection). - TLS/SSL handshake (HTTPS). - Authentication (valid credentials). - API compatibility (server version check).

Successful test output:

Connection Test

```
Server:      https://192.168.1.100:8443
Mode:        Boba Server
Status:      Connected
Version:     1.4.2
Latency:     45ms
Auth:        OK
API:         Compatible
```

If the test fails:

Error	Cause	Solution
Connection refused	Server not running	Start Boba/ qBitTorrent container
Timeout	Firewall or network issue	Check ports, verify network
Certificate error	Self-signed cert not trusted	Accept certificate exception or use trusted cert
Authentication failed	Wrong credentials	Verify API key or username/password

Error	Cause	Solution
API incompatible	Server version too old	Upgrade Boba/qBitTorrent

4.4 Auto-Discover (Optional)

If you don't know your server URL, use auto-discovery:

1. In Options → Server Settings, click **“Discover Servers”**.
2. BobaLink will scan your local network for Boba instances.
3. Discovered servers appear in a list with response times.
4. Click **“Select”** next to your server.
5. Enter credentials and test the connection.

What auto-discovery checks:

URL	Timeout	Purpose
https://boba.local:8443/health	5s	mDNS/bonjour resolution
https://boba.local:8080/health	5s	Alternative port
https://boba:8443/health	5s	Docker hostname
https://localhost:8443/health	3s	Same-machine Docker

5. Configuration Guide

5.1 Boba Server Settings

Setting	Default	Description
Server URL	(empty)	Full HTTPS URL of your Boba server
API Key	(empty)	Authentication key from Boba admin
Connection Timeout	30s	Maximum wait for API responses
Health Check Interval	30s	How often to ping the server

Example configuration:

Server URL: https://boba.home.local:8443
API Key: bob_live_a1b2c3d4e5f6...
Connection Timeout: 30 seconds
Health Check: Every 30 seconds

5.2 qBitTorrent Direct Mode (Without Boba)

Setting	Default	Description
Server URL	(empty)	Full HTTPS URL of qBitTorrent WebUI
Username	(empty)	WebUI username
Password	(empty)	WebUI password
Connection Timeout	30s	Maximum wait for API responses
Health Check Interval	30s	How often to ping the server

Note on Referer header: qBitTorrent WebUI requires the Referer header to match the server origin. BobaLink handles this automatically.

5.3 Authentication Setup

API Key Authentication (Boba)

1. Obtain an API key from your Boba server administrator.
2. In Options → Server Settings, select “**API Key**”.
3. Paste the key into the input field.
4. The key is masked for security. Click the eye icon to reveal.
5. Click “**Test Connection**” to validate.

Username/Password (qBitTorrent)

1. In Options → Server Settings, select “**qBitTorrent Direct**”.
2. Enter your WebUI username.
3. Enter your WebUI password.
4. Click “**Test Connection**”.

Bobalink will automatically: - Authenticate and receive a session cookie. - Store the cookie for subsequent requests. - Refresh the session when it expires.

Basic Authentication (Advanced)

For servers behind reverse proxies with Basic Auth:

1. Select “**Basic Auth**”.
2. Enter the Basic Auth username and password.
3. BobaLink will send Authorization: Basic {base64} headers.

Custom Header (Advanced)

For reverse proxies requiring custom headers (e.g., Cloudflare Access):

1. Select “**Custom Header**”.
2. Enter the header name (e.g., CF-Access-Client-Id).
3. Enter the header value.

5.4 Category Configuration

Categories organize torrents in qBitTorrent. To configure:

1. Open Options → Download Preferences.
2. Enter a default category name (e.g., movies).
3. Categories are created automatically in qBitTorrent 4.5+.
4. For older versions, pre-create categories in qBitTorrent WebUI.

Common category schemes:

Category	Use Case
movies	Film downloads
tv-shows	Television series
music	Audio files
software	Applications and tools
games	Game files
books	E-books and documents
browser	Auto-categorized browser sends

5.5 Download Path Settings

Override the default download directory:

1. Open Options → Download Preferences.
2. Enter an absolute path in **Default Save Path**.

Path examples:

OS	Example Path
Linux	/mnt/downloads/torrents
macOS	/Users/username/Downloads/Torrents
Windows	D:\\Downloads\\Torrents
Docker (Linux container)	/downloads/movies

Important: The path must be valid on the **server** (where qBitTorrent runs), not your local machine.

Using qBitTorrent's Automatic Torrent Management:

1. In qBitTorrent WebUI → Options → Downloads.
2. Enable “**Automatic Management Mode**”.
3. Set default save paths per category.
4. Torrents are automatically moved to category folders on completion.

6. Verification

6.1 Test on a Known Torrent Site

After configuration, verify that BobaLink works:

1. Open a new browser tab.
2. Navigate to a site with known magnet links. For testing, you can use:
 - <https://ubuntu.com/download/alternative-downloads> (official Ubuntu torrents)
 - Any torrent index site
3. Wait for the page to fully load.
4. Look at the BobaLink icon in the toolbar.

Expected behavior: - The badge shows a number (count of detected torrents). - The badge is green (connected to server).

6.2 Verify Detection

1. Click the BobaLink icon to open the popup.
2. You should see a list of detected torrents.

3. Each torrent shows:

- Name (e.g., “Ubuntu 24.04 LTS Desktop”)
- Size (if available)
- Source (magnet link or .torrent file)

If no torrents are detected: - Check that the page actually contains magnet links. - Try scrolling down to trigger lazy loading. - Check the Troubleshooting section.

6.3 Verify Sending

1. In the popup, check one or more torrents.
2. Click **“Send Selected”**.
3. Wait for the confirmation:
 - Browser notification: “Torrent Sent — {name} added to download queue”
 - Or check the popup status bar.

6.4 Check qBitTorrent for Download

1. Open your qBitTorrent WebUI in a browser.
2. Check the transfer list.
3. The sent torrent should appear with status “Downloading” or “Stalled”.
4. If using Boba Server, check the Boba dashboard for job status.

Complete verification checklist:

Check	Method	Expected Result
Extension installed	Icon visible in toolbar	Yes
Server connected	Badge is green	Yes
Detection works	Badge shows count on torrent page	> 0
Send works	Send a test torrent	Success notification
Queue works	Disconnect server, send, reconnect	Item sent after reconnect
Settings persist	Restart browser, check Options	Settings retained

7. Updating

7.1 Automatic Updates from Store

If installed from Chrome Web Store, AMO, or Opera Addons: - Updates are applied automatically by the browser. - No user action required. - Check `chrome://extensions` to see the current version.

7.2 Manual Update Process

For manual installations:

1. Download the latest release ZIP from GitHub.
2. Extract to the same location as the previous installation (overwrite).
3. Navigate to your browser's extensions page.
4. Find BobaLink and click the **refresh/reload** icon.
5. Verify the version number updated.

Chrome:

`chrome://extensions` → Find BobaLink → Click refresh icon

Firefox:

`about:debugging` → This Firefox → Find BobaLink → Click Reload

7.3 Version Compatibility

BobaLink	Boba Server	qBitTorrent	Status
1.0.x	1.0.x – 1.4.x	4.4+	Fully compatible
1.0.x	1.5.x+	4.4+	May have minor issues
1.0.x	< 1.0.x	4.4+	Not supported
1.0.x	Any	< 4.4.0	WebUI API may differ

Before updating: 1. Check the release notes for breaking changes. 2. Back up your configuration (export from Options). 3. Ensure your server versions are compatible.

8. Uninstallation

8.1 Remove from Chrome

Method 1 — Extensions page: 1. Navigate to `chrome://extensions`. 2. Find BobaLink. 3. Click **“Remove”**. 4. Confirm by clicking **“Remove”** again.

Method 2 — Toolbar icon: 1. Right-click the BobaLink icon. 2. Select **“Remove from Chrome”**. 3. Confirm.

8.2 Remove from Firefox

1. Navigate to `about:addons`.
2. Find BobaLink in the Extensions list.
3. Click the `⋮` menu → **“Remove”**.
4. Confirm.

8.3 Remove from Opera

1. Navigate to `opera://extensions`.
2. Find BobaLink.
3. Click **“Remove”**.
4. Confirm.

8.4 Remove from Yandex

1. Navigate to `browser://extensions`.
2. Find BobaLink.
3. Click **“Remove”**.
4. Confirm.

8.5 Clean Up Stored Data

Chrome:

`chrome://settings/cookies/detail?site=bobalink` (if applicable)

All browsers: BobaLink stores data in `chrome.storage.local` which is automatically cleared when the extension is removed. This includes: - Server configuration (URLs, credentials) - Queue items - UI preferences - Encryption keys (session storage, cleared on browser restart)

Manual cleanup (if data persists):

```
// Open DevTools for the extension background page
// Run in Console:
chrome.storage.local.getBytesInUse(null, (bytes) => {
  console.log(`Storage used: ${bytes} bytes`);
});

// Clear all data
chrome.storage.local.clear(() => {
  console.log('All extension data cleared');
});
```

9. Troubleshooting Installation Issues

9.1 “This extension is blocked by the administrator”

Cause: Enterprise policies restrict extension installation.

Solutions:

Option A — Whitelist the extension (IT Admin):

Windows Registry:

HKEY_LOCAL_MACHINE\SOFTWARE\Policies\Google\Chrome\ExtensionInstall
Add the extension ID

Or via Group Policy:

Administrative Templates → Google Chrome → Extensions →
Configure extension installation allow list

Option B — Force install (IT Admin):

Group Policy:

Administrative Templates → Google Chrome → Extensions →
Configure force-installed apps and extensions
Add: {extension-id};https://clients2.google.com/service/update2/
crx

9.2 “Manifest file is missing or unreadable”

Cause: Wrong folder selected when loading unpacked.

Solution: 1. Ensure you extracted the ZIP completely. 2. Select the folder containing `manifest.json`, not a parent or subfolder. 3. Verify `manifest.json` exists: `bash ls /opt/bobalink/chrome/manifest.json`

9.3 “Invalid value for ‘content_security_policy’”

Cause: Browser version too old for MV3 CSP format.

Solution: Upgrade your browser to the minimum required version (Chrome 88+, Firefox 109+).

9.4 Extension Loads But Icon Is Gray

Cause: Extension not configured.

Solution: 1. Click the icon to open the popup. 2. Follow the first-time setup wizard. 3. Configure your server URL and credentials.

9.5 “Failed to load extension from...”

Common causes and solutions:

Error Message	Cause	Solution
Cannot load extension with file or directory name _metadata	macOS metadata folder	Delete __MACOSX folder from extracted ZIP
Invalid locale file	Corrupted _locales	Re-download the ZIP
Required value version is missing	Wrong folder selected	Navigate deeper to find manifest.json
Permission 'XYZ' is unknown	Browser doesn't support a permission	Update browser or use a different build

9.6 Certificate Warnings

When using self-signed certificates (common in home lab setups):

Chrome: 1. Navigate to your server URL (e.g., https://192.168.1.100:8443). 2. Click “**Advanced**” → “**Proceed to {server} (unsafe)**”. 3. The certificate is now trusted for this session.

For permanent trust:

```
# Export the server certificate
openssl s_client -connect 192.168.1.100:8443 </dev/null 2>/dev/
null \
| openssl x509 -outform PEM > boba-cert.pem
```

```
# Install system-wide (Linux)
sudo cp boba-cert.pem /usr/local/share/ca-certificates/boba.crt
sudo update-ca-certificates

# macOS
sudo security add-trusted-cert -d -r trustRoot \
  -k /Library/Keychains/System.keychain boba-cert.pem

# Windows (PowerShell admin)
Import-Certificate -FilePath boba-cert.pem \
  -CertStoreLocation Cert:\LocalMachine\Root
```

9.7 Network Timeouts During Setup

Symptom	Cause	Solution
“Connection timed out”	Firewall blocking port	Open port 8443 (Boba) or 8080 (qBT)
“Network Error”	HTTPS required but HTTP used	Change URL to https://
“DNS resolution failed”	.local domain not resolving	Use IP address instead of hostname
“Connection refused”	Wrong port or service down	Verify port with netstat

9.8 Firefox-Specific Issues

Issue	Solution
“Add-on could not be installed” (corrupt)	The XPI must be signed. Use <code>about:debugging</code> temporary load for unsigned.
chrome.* API errors	The <code>webextension-polyfill</code> may not be loading. Check build output.
Storage quota exceeded	Firefox has stricter storage limits. Clear old queue items.
Service worker not persistent	Firefox suspends service workers more aggressively. Use <code>chrome.alarms</code> to keep alive.

9.9 Docker-Specific Issues

If running Boba or qBitTorrent in Docker:

```
# Verify container is running
```

```
docker ps | grep boba
```

```
# Check container logs
```

```
docker logs boba-container-name
```

```
# Verify port mapping
```

```
docker port boba-container-name
```

```
# Expected: 0.0.0.0:8443->8443/tcp
```

```
# Test from inside the container
```

```
docker exec -it boba-container-name curl localhost:8443/health
```

```
# Check container network
```

```
docker network inspect bridge
```

Common Docker issues:

Issue	Cause	Solution
Port not exposed	Missing -p flag	Add -p 8443:8443 to run command
Container exits immediately	Config error	Check logs with <code>docker logs</code>
Cannot reach from host	Binding to 127.0.0.1 only	Use -p 0.0.0.0:8443:8443
SSL errors	Certificate not mounted	Mount cert volume: -v /path/to/certs:/app/certs

9.10 Getting Help

If you encounter an issue not covered here:

1. Check the [User Guide Troubleshooting](#) section.
2. Check GitHub Issues: <https://github.com/bobaproject/bobalink/issues>
3. Enable debug logging (see [Developer Guide](#)).
4. Collect the following information for support:
 - Browser and version
 - BobaLink version
 - Server type (Boba/qBitTorrent) and version

- Error messages (screenshots or text)
- Steps to reproduce

End of Installation Guide